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Assignment-2

Q. Hadoop Multi-Node Installation on WSL (1 Master + 1 Worker)

Step 1: Export Existing Ubuntu as Master : Command (PowerShell): wsl --export Ubuntu ubuntu-master.tar

Step 2: Create Worker Node

Commands (PowerShell): mkdir C:\wsl\worker1

wsl --import Ubuntu-worker1 C:\wsl\worker1 ubuntu-master.tar

```
PS C:\Users\kanishka> wsl --export Ubuntu ubuntu-master.tar
Export in progress, this may take a few minutes. (4076 MB)

The operation completed successfully.
PS C:\Users\kanishka> mkdir C:\wsl\worker1


    Directory: C:\wsl


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        22-04-2026     14:03                worker1


PS C:\Users\kanishka> wsl --import Ubuntu-worker1 C:\wsl\worker1 ubuntu-master.tar
The operation completed successfully.
PS C:\Users\kanishka>
```

Step 3: Start Master and Worker

Commands:

wsl -d Ubuntu

wsl -d Ubuntu-worker1

Step 4: Set Hostnames

Master: sudo hostnamectl set-hostname master

Worker: sudo hostnamectl set-hostname worker1

Step 5: Update /etc/hosts (Both Nodes)

Add:

127.0.0.1 master

127.0.0.1 worker1

```
Ubuntu                    kanishka@LAPTOP-SBH634N3
kanishka@LAPTOP-SBH634N3:~$ sudo hostnamectl set-hostname master
[sudo] password for kanishka:
kanishka@LAPTOP-SBH634N3:~$ sudo nano /etc/hosts
sudo: unable to resolve host master: Temporary failure in name resolution
kanishka@LAPTOP-SBH634N3:~$ sudo nano /etc/hosts
kanishka@LAPTOP-SBH634N3:~$ hostname
master
kanishka@LAPTOP-SBH634N3:~$
```

```
Ubuntu                    kanishka@LAPTOP-SBH634N:
kanishka@LAPTOP-SBH634N3:~$ sudo hostnamectl set-hostname worker1
[sudo] password for kanishka:
kanishka@LAPTOP-SBH634N3:~$ sudo nano /etc/hosts
sudo: unable to resolve host worker1: Temporary failure in name resolution
kanishka@LAPTOP-SBH634N3:~$ sudo nano /etc/hosts
kanishka@LAPTOP-SBH634N3:~$ hostname
worker1
kanishka@LAPTOP-SBH634N3:~$
```

```
  GNU nano 7.2                          /etc/hosts *
# This file was automatically generated by WSL. To stop automatic generation of thi>
# [network]
# generateHosts = false
127.0.0.1 localhost
127.0.0.1 master
127.0.0.1 worker1

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
127.0.0.1 master
127.0.0.1 worker1
```

Step 6: Configure SSH on Worker

Edit sshd_config: Port 2223 (because mime 2222 is already in use)

```
  GNU nano 7.2                          /etc/ssh/sshd_config
# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2223

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Then:

sudo systemctl stop ssh.socket

sudo systemctl disable ssh.socket

sudo systemctl restart ssh

Step 7: Passwordless SSH (Master to Worker)

ssh-keygen -t rsa

ssh-copy-id -p 2222 username@worker1

```
kanishka@worker1:~$ ssh -p 2223 kanishka@worker1
The authenticity of host '[worker1]:2223 ([127.0.0.1]:2223)' can't be established.
ED25519 key fingerprint is SHA256:r0Mt5fdHrZA+hkZr/fFfC52cHPev+0dX2t2aEwkRRZs.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[worker1]:2223' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Apr 22 09:06:15 UTC 2026

  System load:  0.0                 Processes:             29
  Usage of /:   0.4% of 1006.85GB   Users logged in:       1
  Memory usage: 6%                  IPv4 address for eth0: 172.18.105.91
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Wed Apr 22 09:06:16 2026 from 127.0.0.1
kanishka@worker1:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kanishka/.ssh/id_rsa):
/home/kanishka/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
```

```
kanishka@worker1:~$ ssh-copy-id -p 2223 kanishka@worker1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kanishka/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are alre
ady installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to insta
ll the new keys
kanishka@worker1's password:
Permission denied, please try again.
kanishka@worker1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh -p 2223 'kanishka@worker1'"
and check to make sure that only the key(s) you wanted were added.
```

Step 8: Hadoop Configuration (Master Only)

Edit core-site.xml, hdfs-site.xml, yarn-site.xml, mapred-site.xml

Set fs.defaultFS to hdfs://master:9000

Set dfs.replication to 1

```
kanishka@worker1:~$ cd /usr/local/hadoop/etc/hadoop
kanishka@worker1:/usr/local/hadoop/etc/hadoop$ nano core-site.xml
kanishka@worker1:/usr/local/hadoop/etc/hadoop$ nano hdfs-site.xml
kanishka@worker1:/usr/local/hadoop/etc/hadoop$ mkdir -p /usr/local/hadoop/data/namenode
kanishka@worker1:/usr/local/hadoop/etc/hadoop$ mkdir -p /usr/local/hadoop/data/datanode
kanishka@worker1:/usr/local/hadoop/etc/hadoop$ hdfs namenode -format
2026-04-22 09:16:26,924 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = worker1/127.0.0.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.6
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/re2
```

Then, Hadoop by default uses **port 22**, but your worker is on **2223.**

So edit:

nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh

Add this line:

export HADOOP_SSH_OPTS="-p 2223 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout=10s"

```
  GNU nano 7.2                  /usr/local/hadoop/etc/hadoop/hadoop-env.sh
###

# There are some optional components of hadoop that allow for
# command and control of remote hosts.  For example,
# start-dfs.sh will attempt to bring up all NNs, DNS, etc.

# Options to pass to SSH when one of the "log into a host and
# start/stop daemons" scripts is executed
export HADOOP_SSH_OPTS="-p 2223 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout=10s"


^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Step 9: Create Hadoop Data Directories

Master:

/usr/local/hadoop/data/namenode

/usr/local/hadoop/data/datanode

Worker:

/usr/local/hadoop/data/datanode

Step 10: Format NameNode

Command: hdfs namenode -format

Step 11: Start Hadoop

Commands:

start-dfs.sh
start-yarn.sh

Step 12: Verification

Commands:

jps
hdfs dfsadmin -report

Step 13: HDFS Test

Commands:

hdfs dfs -mkdir /demo
hdfs dfs -put /etc/passwd /demo
hdfs dfs -ls /demo

```
kanishka@worker1:~$ nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
kanishka@worker1:~$ start-dfs.sh
Starting namenodes on [localhost]
localhost: kanishka@localhost: Permission denied (publickey,password).
Starting datanodes
worker1: kanishka@worker1: Permission denied (publickey,password).
Starting secondary namenodes [worker1]
worker1: kanishka@worker1: Permission denied (publickey,password).
kanishka@worker1:~$ nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
kanishka@worker1:~$ source ~/.bashrc
kanishka@worker1:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [worker1]
kanishka@worker1:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 2174.  Stop it first and ensure /tmp/hadoop-kanishka-resourcema
nager.pid file is empty before retry.
Starting nodemanagers
kanishka@worker1:~$ jps
3136 DataNode
3665 NodeManager
3366 SecondaryNameNode
3015 NameNode
3835 Jps
2174 ResourceManager
kanishka@worker1:~$ hdfs dfsadmin -report
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021697179648 (951.53 GB)
DFS Remaining: 1021697155072 (951.53 GB)
DFS Used: 24576 (24 KB)
DFS Used%: 0.00%
Replicated Blocks:
        Under replicated blocks: 0
        Blocks with corrupt replicas: 0
        Missing blocks: 0
        Missing blocks (with replication factor 1): 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0
```

```
Erasure Coded Block Groups:
        Low redundancy block groups: 0
        Block groups with corrupt internal blocks: 0
        Missing block groups: 0
        Low redundancy blocks with highest priority to recover: 0
        Pending deletion blocks: 0

-------------------------------------------------
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: localhost
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 4411641856 (4.11 GB)
DFS Remaining: 1021697155072 (951.53 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.51%
```

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```
DFS Remaining%: 94.51%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Wed Apr 22 09:25:21 UTC 2026
Last Block Report: Wed Apr 22 09:24:11 UTC 2026
Num of Blocks: 0


kanishka@worker1:~$ hdfs dfs -mkdir /demo
kanishka@worker1:~$ hdfs dfs -put /etc/passwd /demo
kanishka@worker1:~$ hdfs dfs -ls /demo
Found 1 items
-rw-r--r--   1 kanishka supergroup       1476 2026-04-22 09:26 /demo/passwd
kanishka@worker1:~$
```

Namenode information | All Applications | Ask Gemini

localhost:9870/dfshealth.html#tab-overview

3 files and directories, 1 blocks (1 replicated blocks, 0 erasure coded block groups) = 4 total filesystem object(s).

Heap Memory used 170.04 MB of 308 MB Heap Memory. Max Heap Memory is 1.68 GB.

Non Heap Memory used 52.31 MB of 54.19 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	1006.85 GB
Configured Remote Capacity:	0 B
DFS Used:	25.46 KB (0%)
Non DFS Used:	4.11 GB
DFS Remaining:	951.53 GB (94.51%)
Block Pool Used:	25.46 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0
Total Datanode Volume Failures	0 (0 B)

Block Deletion Start Time	Wed Apr 22 14:54:07 +0530 2026
Last Checkpoint Time	Wed Apr 22 14:46:28 +0530 2026
Enabled Erasure Coding Policies	RS-6-3-1024k

NameNode Journal Status

Current transaction ID: 8

Journal Manager	State
FileJournalManager(root=/tmp/hadoop-kanishka/dfs/name)	EditLogFileOutputStream(/tmp/hadoop-kanishka/dfs/name/current/edits_inprogress_0000000000000000001)

NameNode Storage

Storage Directory	Type	State
/tmp/hadoop-kanishka/dfs/name	IMAGE_AND_EDITS	Active

DFS Storage Types

Storage Type	Configured Capacity	Capacity Used	Capacity Remaining	Block Pool Used	Nodes In Service
DISK	1006.85 GB	25.46 KB (0%)	951.53 GB (94.51%)	25.46 KB	1

Hadoop, 2023.